25.2: The fossil record documents the history of life

Starting with the earliest traces of life, the fossil record opens a window into the world of long ago and provides glimpses of the evolution of life over billions of years. In this section, we'll examine fossils as a form of scientific evidence: how fossils form, how scientists date and interpret them, and what they can and cannot tell us about changes in the history of life.

The Fossil Record

Sedimentary rocks are the richest source of fossils. As a result, the fossil record is based primarily on the sequence in which fossils have accumulated in sedimentary rock layers, called *strata* (see Figure 22.3). Useful information is also provided by other types of fossils, such as insects preserved in amber (fossilized tree sap) and mammals frozen in soil **(Figure 25.5)**.

Figure 25.5

A gallery of fossil types.

A gallery of fossil types.

(a) Sedimentary rock. *Most fossils are found in sedimentary rocks, such as this plant fossil.* ***(b) Mineralized organic matter.*** *Some fossils, such as this petrified tree trunk, form as minerals seep into and replace organic matter.* ***(c) Trace fossils.*** *Footprints, burrows, or other traces of an organism's activities can be preserved in the fossil record.* ***(d) Amber.*** *Entire organisms are sometimes found preserved in hardened resin from a tree.* ***(e) Frozen soil, ice, and acid bogs.*** *Rarely, frozen soil, ice, or an acid bog can preserve the body of larger organisms, such as this wolf pup.*

Figure a is a plant fossil that is about 300 million years old. The outline of a plant is visible in rocks.

Figure b is a petrified tree in Arizona. It is about 215 million years old.

Figure c is 150 million-year-old dinosaur tracks in Colorado. A child or small adult stands in one of the tracks with dirt reaching their thighs.

Figure d is an insect in amber. It appears to be a mosquito or mosquito-like insect. It is about 40 million years old.

Figure e is a 50,000 year-old wolf pup found in frozen soil in Yukon, Canada.

The fossil record shows that there have been great changes in the kinds of organisms on Earth at different points in time. Many past organisms were unlike organisms living today, and many organisms that once were common are now extinct. As we'll see later in this section, fossils also document how new groups of organisms arose from previously existing ones.

As substantial and significant as the fossil record is, keep in mind that it is an incomplete chronicle of evolution. Many of Earth's organisms did not die in the right place and time to be preserved as fossils. Of the fossils that were formed, many were destroyed by later geologic processes, and only a fraction of the others have been discovered. As a result, the known fossil record is biased in favor of species that existed for a long time, were abundant and widespread in certain kinds of environments, and had hard shells, skeletons, or other parts that facilitated their fossilization. Even with its limitations, however, the fossil record is a remarkably detailed account of biological change over the vast scale of geologic time. Furthermore, as shown by recently unearthed fossils of whale ancestors with hind limbs (see Figure 22.19 and Figure 25.1), gaps in the fossil record continue to be filled by new discoveries.

How Rocks and Fossils Are Dated

Fossils provide valuable data for reconstructing the history of life, but only if we can determine where they fit in that story. While the order of fossils in rock strata tells us the sequence in which the fossils were laid down—their relative ages—it does not tell us their actual ages. Examining the relative positions of fossils is like peeling off layers of wallpaper in an old house. You can infer the sequence in which the layers were applied, but not the year each layer was added.

How can we determine the age of a fossil? One of the most common techniques is ******radiometric dating**, which is based on the decay of radioactive isotopes (see Concept 2.2). In this process, a radioactive "parent" isotope decays to a "daughter" isotope at a characteristic rate. The rate of decay is expressed by the ******half-life**, the time required for 50% of the parent isotope to decay **(Figure 25.6)**. Each type of radioactive isotope has a characteristic half-life, which is not affected by temperature, pressure, or other environmental variables. For example, carbon-14 decays relatively quickly; its half-life is 5,730 years. Uranium-238 decays slowly; its half-life is 4.5 billion years.

Figure 25.6

Radiometric dating.

A graph plots fraction of parent isotope remaining versus time in half-lives.

In this diagram, each unit of time represents one half-life of a radioactive isotope.

A smooth curve falls through (0, y), (1, one-half), (2, one-fourth), (3, one-eighth), and (4, one-sixteenth). The vertical distance between the starting point and the curve is labeled accumulating daughter isotope. The vertical distance between the curve and the x-axis is labeled remaining parent isotope.

DRAW IT *Relabel the*x-axis*of this graph in years to illustrate the radioactive decay of*uranium-238** (half-life = 4.5 billion years).

*For suggested answer, see*Appendix A*.*

Fossils contain isotopes of elements that accumulated in the organisms when they were alive. For example, a living organism contains the most common carbon isotope, carbon-12, as well as a radioactive isotope, carbon-14. When the organism dies, it stops accumulating carbon, and the amount of carbon-12 in its tissues does not change over time. However, the carbon-14 that it contains at the time of death slowly decays into another element, nitrogen-14. Thus, by measuring the ratio of carbon-14 to carbon-12 in a fossil, we can determine the fossil's age. This method works for fossils up to about 75,000 years old; fossils older than that contain too little carbon-14 to be detected with current techniques. Radioactive isotopes with longer half-lives are used to date older fossils.

Determining the age of these older fossils in sedimentary rocks can be challenging. Organisms do not use radioisotopes with long half-lives, such as uranium-238, to build their bones or shells. In addition, sedimentary rocks are often composed of sediments of differing ages. Although we cannot date these older fossils directly, an indirect method can be used to infer the age of fossils that are sandwiched between two layers of volcanic rock. As lava cools into volcanic rock, radioisotopes from the surrounding environment become trapped in the newly formed rock. Some of the trapped radioisotopes have long half-lives, allowing geologists to estimate the ages of ancient volcanic rocks. If two volcanic layers surrounding fossils are found to be 525 million and 535 million years old, for example, then the fossils are roughly 530 million years old.

The study of fossils has helped geologists establish a **geologic record**: a standard time scale that divides Earth's history into four eons and further subdivisions **(Table 25.1)**. We'll describe key events that occurred during these different time periods in Concept 25.3.

Table 25.1

The Geologic Record

A table lists the breakdown of time as eons, eras, periods, epochs, and ages (millions of years ago).

The table is as follows.

Eon	Era	Period	Epoch	Age (millions of years ago)

Phanerozoic	Cenozoic	Quaternary	Holocene	0.01
Phanerozoic	Cenozoic	Quaternary	Pleistocene	2.6
Phanerozoic	Cenozoic	Neogene	Pliocene	5.3
Phanerozoic	Cenozoic	Neogene	Miocene	23
Phanerozoic	Cenozoic	Paleogene	Oligocene	33.9
Phanerozoic	Cenozoic	Paleogene	Eocene	56
Phanerozoic	Cenozoic	Paleogene	Paleocene	66
Phanerozoic	Mesozoic	Cretaceous	blank	145
Phanerozoic	Mesozoic	Jurassic	blank	201
Phanerozoic	Mesozoic	Triassic	blank	252
Phanerozoic	Paleozoic	Pemeian	blank	299
Phanerozoic	Paleozoic	Carboniferous	blank	359
Phanerozoic	Paleozoic	Devononian	blank	419
Phanerozoic	Paleozoic	Silurian	blank	444
Phanerozoic	Paleozoic	Ordovician	blank	485
Phanerozoic	Paleozoic	Cambrian	blank	541
Phanerozoic	Neo-Proterozoic	Ediacaran	blank	635
Phanerozoic	Neo-Proterozoic	blank	blank	1000
Proterozoic	Neo-Proterozoic	blank	blank	2500
Archaean	blank	blank	blank	4000
Hadean	blank	blank	blank	Approximately 4600

The Origin of New Groups of Organisms

Some fossils document the origin of new groups of organisms. Such fossils are central to our understanding of evolution; they illustrate how new features arise and how long it takes for such changes to occur. We'll

examine one such case here and in **Figure 25.7**: the origin of mammals.

Figure 25.7

Exploring the Origin of Mammals

An evolutionary tree diagram gives the lineage of the mammals from a common ancestor with other tetrapods.

Parts are as follows.

Over the course of 120 million years, mammals originated gradually from a group of tetrapods called synapsids. Shown here are a few of the many fossil organisms whose morphological features represent intermediate steps between living mammals and their early synapsid ancestors. The evolutionary context of the origin of mammals is shown in the tree diagram at right (the dagger symbol, looks like a dagger pointing downward, indicates extinct lineages). Key to the color coding of skull bones in this graphic: orange = articular, tan = dentary, green = quadrate, purple = squamosal.

Tree diagram showing lineage of the mammals from a common ancestor with other tetrapods. Tree is as follows.

Other tetrapods; Branch point 1 Reptiles (including dinosaurs and birds; example is snake). Synapsids; Branch point 2. Dimetrodon (extinct; example shows reptile like animal with large sail on back). Therapsids; Cynodonts; Branch point 3. Very late (non-mammalian) cynodonts (example is animal that resembles a mammal); extinct. Mammals (example is a black bear)

Early synapsids had multiple bones in the lower jaw and single-pointed teeth. The jaw hinge was formed by the articular and quadrate bones. Early synapsids also had an opening called the temporal fenestra behind the eye socket. Powerful cheek muscles for closing the jaws probably passed through the temporal fenestra. Over time, this opening enlarged and moved in front of the hinge between the lower and upper jaws, thereby increasing the power and precision with which the jaws could be closed (much as moving a doorknob away from the hinge makes a door easier to close).

Synapsid skull with sharp teeth. Temporal fenestra (opening) behind eye socket; squamosal bone just below; quadrate bone just below squamosal. Below quadrate is jaw hinge; just below that is the articular bone. The dentary bone extends from the front of the lower jaw back and does not reach the hinge.

Later, a group of synapsids called therapsids appeared. Therapsids had large dentary bones, long faces, and the first examples of specialized teeth, large canines. These trends continued in a group of therapsids called cynodonts. Therapsid skull with prominent canines. Temporal fenestra opening is just behind eye socket, but is longer and thinner than in the synapsids. Squamosal bone surrounds temporal fenestra; quadrate directly below; then hinge, followed by articular bone just below hinge. All four bones (squamosal, quadrate,

articular and dentary) have a slightly different shape than they do in the synapsids. Temporal fenestra is thinner; hinge is smaller and more posterior, articular bends in the opposite direction, dentary is larger. Overall skull is flatter and longer.

In early cynodont therapsids, the dentary was the largest bone in the lower jaw, the temporal fenestra was large and positioned forward of the jaw hinge, and teeth with several cusps first appeared (not visible in the diagram). As in earlier synapsids, the jaw had an articular-quadrate hinge. Early cynodont skull with temporal fenestra, squamosal, quadrate, articular and dentary bones. Temporal fenestra (opening) is much smaller than in previous skulls and hardly visible. Squamosal bone is no longer adjacent to the temporal fenestra and is separated by an expanse of skull. Quadrate remains below squamosal just above hinge but is thicker than in the therapsids; articular bone is below hinge. Dentary bone extends further back along the lower jaw than in the therapsids. Skull is flatter than in the therapsids.

Later cynodonts had teeth with complex cusp patterns, and their lower and upper jaws hinged in two locations: They retained the original articular-quadrate hinge and formed a new, second hinge between the dentary and squamosal bones. (The temporal fenestra is not visible in this or the below cynodont skull at the angles shown.)

In some very late (nonmammalian) cynodonts and early mammals, the original articular-quadrate hinge was lost, leaving the dentary-squamosal hinge as the only hinge between the lower and upper jaws, as in living mammals. The articular and quadrate bones migrated into the ear region (not shown), where they functioned in transmitting sound. In the mammal lineage, these two bones later evolved into the familiar hammer (malleus) and anvil (incus) bones of the ear.

Along with amphibians and reptiles, mammals belong to the group of animals called *tetrapods* (from the Greek *tetra*, four, and *pod*, foot), named for having four limbs. Mammals have a number of unique anatomical features that fossilize readily, allowing scientists to trace their origin. For example, the lower jaw is composed of one bone (the dentary) in mammals but several bones in other tetrapods. In addition, the lower and upper jaws in mammals hinge between a different set of bones than in other tetrapods. Mammals also have a unique set of three bones that transmit sound in the middle ear, the hammer, anvil, and stirrup, whereas other tetrapods have only one such bone, the stirrup (see Concept 34.6). Finally, the teeth of mammals are differentiated into incisors (for tearing), canines (for piercing), and the multi-pointed premolars and molars (for crushing and grinding). In contrast, the teeth of other tetrapods usually consist of a row of undifferentiated, single-pointed teeth.

As detailed in Figure 25.7, the fossil record shows that the unique features of mammalian jaws and teeth evolved gradually, in a series of steps. Keep in mind that the figure includes just a few examples of the fossil skulls that document the origin of mammals. If all the known fossils in the sequence were arranged by shape and placed side by side, their features would blend smoothly from one group to the next. Some of these

fossils would reflect how the features of a group that dominates life today, the mammals, gradually arose in a previously existing group, the cynodonts. Others would reveal side branches on the tree of life—groups of organisms that thrived for millions of years but ultimately left no descendants that survive today.